CLOUD COMPANY

SECURE NETWORK DESIGN

Final Project Documentation

Enterprise Network Infrastructure Design & Implementation

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1. Executive Summary

Cloud Company is a rapidly growing cloud solutions provider serving clients worldwide. Due to significant business expansion, the company has relocated to a new three-floor office building accommodating approximately 600 employees across multiple departments.

This document presents the final technical documentation for the design and implementation of Cloud Company's new enterprise network infrastructure. The network has been engineered to deliver high performance, scalability, redundancy, and robust security while supporting both internal and external communications — including wireless connectivity, VoIP services, cloud access, and dual-ISP Internet redundancy.

All components have been configured, integrated, and verified through comprehensive testing. This report covers the project scope, topology, IP addressing, device configurations, security architecture, and test results.

2. Project Scope & Objectives

2.1 Organizational Overview

The Cloud Company campus spans three floors and hosts the following departments:

- Sales and Marketing
- Human Resources and Logistics
- Finance and Accounts
- Administration and Public Relations
- ICT Department (Software Developers, Cloud Engineers, Cybersecurity Engineers, Network Engineers, System Administrators, IT Support Specialists, Business Analysts, Project Managers)
- Server Room

2.2 Project Objectives

The primary goal was to design and implement a secure, scalable, and highly available enterprise network infrastructure. The specific objectives included:

- Implement a hierarchical enterprise network design with Distribution and Access layers
- Establish redundant dual-ISP Internet connectivity (SEACOM & Safaricom)
- Deploy Cisco ASA firewalls with Inside, Outside, and DMZ security zones
- Implement VLAN segmentation across all floors and departments
- Configure HSRP for gateway redundancy and failover
- Deploy Wireless LAN Controller (WLC) with Lightweight Access Points (LAPs)
- Implement VoIP services via Cisco Voice Gateway
- Configure OSPF dynamic routing throughout the enterprise
- Establish EtherChannel (LACP) for link aggregation between core switches
- Deploy DHCP, DNS, Active Directory, and RADIUS services in protected zones
- Host public-facing services (Web, FTP, Email, Application, NAS) in the DMZ

3. Network Topology

3.1 Topology Diagram

The following diagram illustrates the complete network topology as designed and implemented in Cisco Packet Tracer:

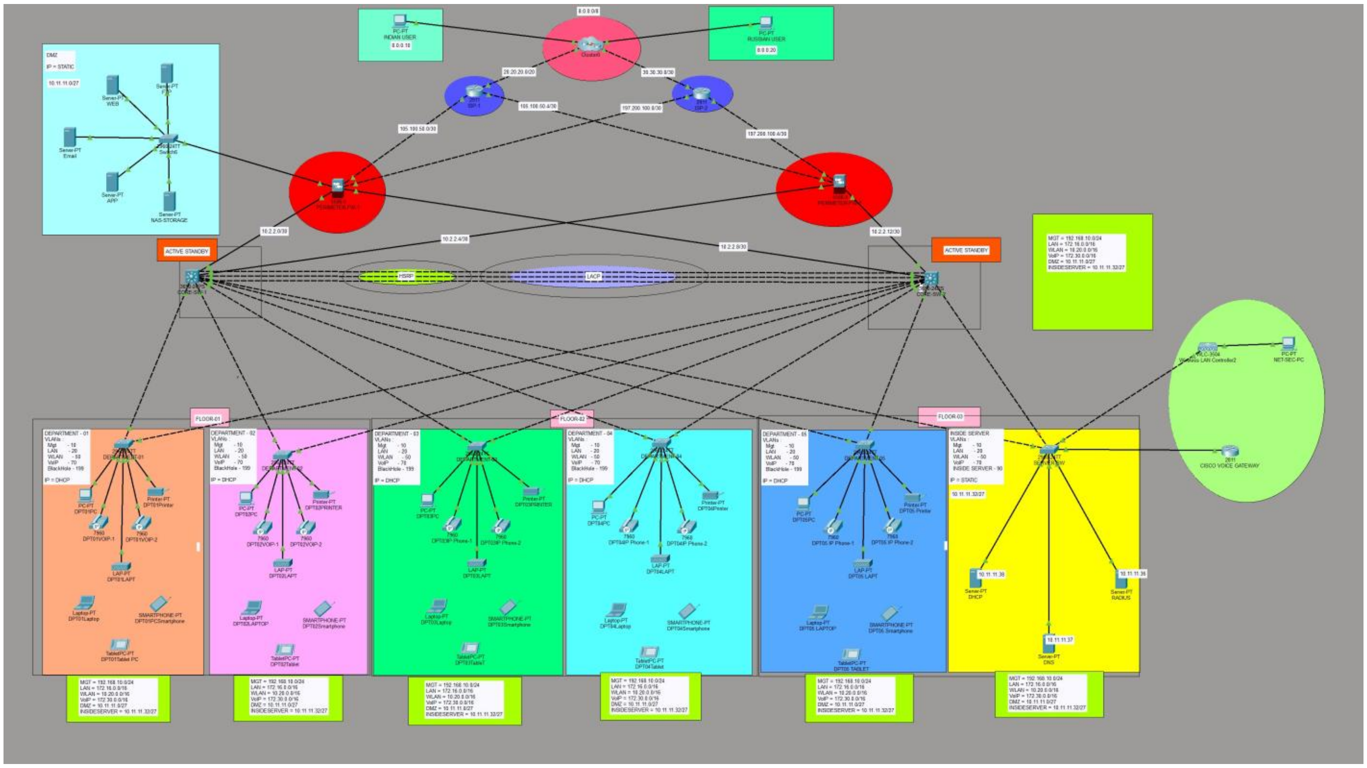

Figure 1: Cloud Company — Complete Enterprise Network Topology

3.2 Hierarchical Design

The network follows a three-tier hierarchical design model:

Layer	Devices & Function
Core / Distribution	Two Cisco Catalyst 3850 Core Multilayer Switches (Core-SW1 & Core-SW2) providing inter-VLAN routing, HSRP, OSPF, and EtherChannel aggregation
Access Layer	Cisco Catalyst 2960 Access Switches per department (one per floor/department), connecting end-user devices
Security Layer	Two Cisco ASA 5500-X Firewalls (FW1 & FW2) forming the security perimeter between Outside, Inside, and DMZ zones
ISP / WAN	Dual-ISP routers connecting to SEACOM (ISP-1) and Safaricom (ISP-2) for redundant Internet access

3.3 Security Zones

Three distinct security zones are enforced by the Cisco ASA firewalls:

- Outside Zone (Security Level 0): Internet-facing interfaces connected to ISP-1 and ISP-2
- DMZ Zone (Security Level 70): Public-facing servers — Web, FTP, Email, Application, NAS Storage
- Inside Zone (Security Level 100): All internal infrastructure — LAN, WLAN, VoIP, Management, and Inside Servers

4. IP Address Allocation

4.1 Network Subnets

Network / VLAN	Address Range	Purpose
VLAN 10 — Management	192.168.10.0/24	Network device management access
VLAN 20 — LAN	172.16.0.0/16	Wired employee workstations
VLAN 50 — WLAN	10.20.0.0/16	Wireless devices (laptops, smartphones, tablets)
VLAN 70 — VoIP	172.30.0.0/16	IP phones and voice traffic
VLAN 199 — Blackhole	N/A	Security VLAN for unused/unassigned ports
VLAN 90 — Inside Server	10.11.11.32/27	Internal servers (AD, DHCP, DNS, RADIUS)
DMZ Network	10.11.11.0/27	DMZ servers (Web, FTP, Email, App, NAS)
ISP-1 (SEACOM)	105.100.50.0/30	WAN link to ISP-1
ISP-2 (Safaricom)	197.200.100.0/30	WAN link to ISP-2
Core Link 1	10.2.2.0/30	FW1 ↔ Core-SW1
Core Link 2	10.2.2.4/30	FW1 ↔ Core-SW2
Core Link 3	10.2.2.8/30	FW2 ↔ Core-SW1
Core Link 4	10.2.2.12/30	FW2 ↔ Core-SW2

4.2 Server Static Addresses

Server	IP Address	Zone
DHCP Server	10.11.11.38	Inside (VLAN 90)
DNS Server	10.11.11.37	Inside (VLAN 90)
RADIUS Server	10.11.11.36	Inside (VLAN 90)
Web Server	10.11.11.x (static)	DMZ
FTP Server	10.11.11.x (static)	DMZ
Email Server	10.11.11.x (static)	DMZ

Server	IP Address	Zone
Application Server	10.11.11.x (static)	DMZ
NAS Storage	10.11.11.x (static)	DMZ

5. VLAN Configuration

5.1 VLAN Table

VLAN ID	Name	Subnet	Purpose
10	MGT	192.168.10.0/24	Management (SSH, monitoring)
20	LAN	172.16.0.0/16	Wired workstations
50	WLAN	10.20.0.0/16	Wireless clients
70	VoIP	172.30.0.0/16	IP telephony
90	INSIDESERVER	10.11.11.32/27	Internal servers
199	Blackhole	—	Security: unassigned ports

5.2 Core Switch VLAN Verification

The following output was captured from Core-SW1 confirming active VLANs:

```
VLAN Name           Status
---- -------------- ------
10   MGT             active
20   LAN             active
50   WLAN            active
70   VoIP            active
90   INSIDESERVER    active
199  Blackhole        active
```

6. Core Switch Configuration

6.1 Core-SW1 — Key Configuration

6.1.1 Interface Configuration

Core-SW1 uplinks to both firewalls via routed interfaces, and connects to access switches via trunk ports:

```
interface GigabitEthernet1/0/1
 no switchport
 ip address 10.2.2.1 255.255.255.252   ! Uplink to FW1 Inside1
```

```
interface GigabitEthernet1/0/8
 no switchport
 ip address 10.2.2.5 255.255.255.252   ! Uplink to FW1 Inside2

interface GigabitEthernet1/0/2 - 1/0/7
 switchport mode trunk                  ! Downlinks to Access Switches

interface GigabitEthernet1/0/9 - 1/0/11
 switchport mode trunk
 channel-group 1 mode active            ! LACP EtherChannel to Core-SW2
```

6.1.2 SVI (Inter-VLAN Routing) Interfaces

```
interface Vlan10
 ip address 192.168.10.3 255.255.255.0
 ip helper-address 10.11.11.38          ! DHCP relay to DHCP server
 standby 10 ip 192.168.10.1             ! HSRP Virtual Gateway

interface Vlan20
 ip address 172.16.0.3 255.255.0.0
 ip helper-address 10.11.11.38
 standby 20 ip 172.16.0.1

interface Vlan50
 ip address 10.20.0.2 255.255.0.0
 ip helper-address 10.11.11.38
 standby 50 ip 10.20.0.1

interface Vlan90
 ip address 10.11.11.34 255.255.255.224
 standby 90 ip 10.11.11.33
```

6.1.3 OSPF Configuration

```
router ospf 35
 router-id 1.1.1.1
 network 10.2.2.0 0.0.0.3 area 0
 network 10.2.2.4 0.0.0.3 area 0
 network 192.168.10.0 0.0.0.255 area 0
 network 172.16.0.0 0.0.255.255 area 0
 network 10.20.0.0 0.0.255.255 area 0
 network 10.11.11.32 0.0.0.31 area 0
```

6.1.4 Security Configuration

```
! SSH Access Control - Only Management VLAN allowed
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 deny any

line vty 0 4
 access-class 1 in
 login local
 transport input ssh

banner motd ^C NO UNAUTHORIZED ACCESS ^C
```

7. HSRP — Gateway Redundancy

7.1 HSRP Overview

Hot Standby Router Protocol (HSRP) has been configured on both Core switches to provide transparent gateway failover for all VLANs. If the Active gateway fails, the Standby switch automatically takes over with zero manual intervention.

7.2 HSRP State — Core-SW1

```
Interface  Grp  Pri P State    Active          Standby         Virtual IP
Vl10       10   100   Active   local           192.168.10.2    192.168.10.1
Vl20       20   100   Active   local           172.16.0.2      172.16.0.1
Vl50       50   100   Standby  10.20.0.3       local           10.20.0.1
Vl90       90   100   Standby  10.11.11.35     local           10.11.11.33
```

Core-SW1 is the Active gateway for VLANs 10 and 20. Core-SW2 is the Active gateway for VLANs 50 and 90, providing load balancing across both core switches.

8. EtherChannel — LACP Link Aggregation

8.1 Configuration

EtherChannel using LACP (IEEE 802.3ad) has been configured between Core-SW1 and Core-SW2 across three GigabitEthernet ports, forming a single logical Port-Channel 1 with 3 Gbps of aggregate bandwidth.

8.2 EtherChannel Verification

```
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------
1      Po1(SU)      LACP        Gig1/0/9(P) Gig1/0/10(P) Gig1/0/11(P)

Flags: SU = Layer2, in use | P = bundled in port-channel
```

All three physical ports are active (P) and the Port-Channel is in use (SU), confirming successful LACP negotiation.

9. OSPF Dynamic Routing

9.1 OSPF Design

OSPF Process 35, Area 0 (Backbone) is deployed throughout the entire enterprise — from ISP routers through both firewalls, both core switches, and all connected networks. This ensures fully dynamic routing with automatic convergence on link failure.

9.2 OSPF Neighbor Table — Core-SW1

```
Neighbor ID    Pri   State          Dead Time   Address          Interface
1.1.8.8         1    FULL/BDR       00:00:30    10.2.2.2         GigabitEthernet1/0/1
1.1.2.2         1    FULL/BDR       00:00:38    192.168.10.2     Vlan10
1.1.2.2         1    FULL/DR        00:00:36    172.16.0.2       Vlan20
1.1.2.2         1    FULL/DR        00:00:39    10.11.11.35      Vlan90
1.1.2.2         1    FULL/DR        00:00:38    10.20.0.3        Vlan50
1.1.9.9         1    FULL/BDR       00:00:39    10.2.2.6         GigabitEthernet1/0/8
```

All OSPF neighbors are in FULL state, confirming complete routing adjacency across the enterprise. Router IDs 1.1.8.8 and 1.1.9.9 correspond to the two ASA Firewalls; 1.1.2.2 is Core-SW2.

10. Cisco ASA Firewall Configuration

10.1 Firewall Architecture

Two Cisco ASA 5500-X Series Firewalls (FW1 & FW2) form the security perimeter. Each firewall connects to both ISPs (outside), both core switches (inside), and the DMZ segment.

10.2 FW1 — Interface Configuration

Interface	Name (Zone)	Security Level	IP Address
GigabitEthernet1/1	outSIDE1	0	105.100.50.2/30
GigabitEthernet1/2	outSIDE2	0	197.200.100.2/30
GigabitEthernet1/3	INSIDE1	100	10.2.2.2/30
GigabitEthernet1/4	INSIDE2	100	10.2.2.10/30
GigabitEthernet1/5	DMZ	70	10.11.11.1/27

10.3 NAT Configuration

Dynamic PAT (Port Address Translation) is configured to allow internal and DMZ hosts to access the Internet through the public ISP interfaces:

```
! DMZ to Internet
object network DMZ-OUTSIDE1
 subnet 10.11.11.0 255.255.255.224
 nat (DMZ,outSIDE1) dynamic interface

! Internal LAN to Internet (Primary ISP)
object network INSIDE1-OUTSIDE1
 subnet 172.16.0.0 255.255.0.0
 nat (INSIDE1,outSIDE1) dynamic interface

! WLAN to Internet
object network INSIDEw1-OUTSIDEw1
```

```
 subnet 10.20.0.0 255.255.0.0
 nat (INSIDE1,outSIDE1) dynamic interface
```

10.4 Static Routing & Failover

```
! Primary route via ISP-1 (SEACOM) - lower metric
route outSIDE1 0.0.0.0 0.0.0.0 105.100.50.1 1

! Backup route via ISP-2 (Safaricom) - higher metric (AD 70)
route outSIDE2 0.0.0.0 0.0.0.0 197.200.100.1 70
```

Traffic flows through ISP-1 by default (metric 1). If ISP-1 fails, traffic automatically fails over to ISP-2 (metric 70), ensuring continuous Internet availability.

10.5 Access Control Lists

```
! ACL: RES — Applied to DMZ and Outside interfaces
access-list RES extended permit icmp any any
access-list RES extended permit tcp any any eq www    ! HTTP
access-list RES extended permit tcp any any eq domain ! DNS TCP
access-list RES extended permit udp any any eq domain ! DNS UDP

access-group RES in interface DMZ
access-group RES in interface outSIDE1
access-group RES in interface outSIDE2
```

10.6 OSPF on ASA

```
router ospf 35
 router-id 1.1.8.8
 network 105.100.50.0 255.255.255.252 area 0
 network 197.200.100.0 255.255.255.252 area 0
 network 10.11.11.0 255.255.255.224 area 0
 network 10.2.2.0 255.255.255.252 area 0
 network 10.2.2.8 255.255.255.252 area 0
```

11. ISP Router — Routing Table

11.1 ISP-1 (SEACOM) Router — Verified Route Table

The following routing table was captured from the ISP-1 router, confirming full reachability of all enterprise subnets via OSPF:

```
O  8.0.0.0/8       via 105.100.50.6 (ISP-2 simulated Internet)
O  10.2.2.0/30    via 105.100.50.2 (FW1 INSIDE1)
O  10.2.2.4/30    via 105.100.50.6
O  10.2.2.8/30    via 105.100.50.2 (FW1 INSIDE2)
O  10.11.11.0/27  via 105.100.50.2 (DMZ)
O  10.11.11.32/27 via 105.100.50.2 & 105.100.50.6 (Inside Servers, dual-path)
O  10.20.0.0/16   via 105.100.50.2 & 105.100.50.6 (WLAN, dual-path)
```

```
O  172.16.0.0/16  via 105.100.50.2 & 105.100.50.6 (LAN, dual-path)
O  192.168.10.0/24 via 105.100.50.2 & 105.100.50.6 (Management, dual-path)
O  197.200.100.0/30 via 105.100.50.6 (ISP-2 WAN link)
C  105.100.50.0/30 directly connected (ISP-1 to FW1)
C  20.20.20.0/30  directly connected (ISP-1 to simulated Internet)
```

ECMP (Equal-Cost Multi-Path) is visible for several enterprise subnets, confirming that the dual-firewall design provides both redundancy and load sharing at the routing level.

12. Wireless & VoIP Infrastructure

12.1 Wireless LAN

A Cisco Wireless LAN Controller (WLC) provides centralized management of all Lightweight Access Points (LAPs) deployed across the three floors. All wireless clients receive IP addresses from VLAN 50 (10.20.0.0/16) via DHCP relay.

- WLC manages LAP registration, RF channel assignment, and roaming
- SSIDs are mapped to VLAN 50 (WLAN) for all wireless traffic
- RADIUS server (10.11.11.36) provides 802.1X authentication for enterprise wireless security
- Each department floor has at least one LAP (DPT01LAPT through DPT06LAPT)

12.2 VoIP Infrastructure

A Cisco Voice Gateway provides IP telephony services to all departments. VoIP traffic is segregated on VLAN 70 (172.30.0.0/16) for QoS prioritization.

- IP phones deployed per department: DPT01VOIP-1 & 2 through DPT05 IP Phone-1 & 2
- QoS policies applied at access layer to prioritize voice traffic
- Voice Gateway integrated with OSPF for routing convergence

13. Security Design Summary

13.1 Defense-in-Depth Architecture

Security Control	Implementation
Perimeter Firewall	Dual Cisco ASA 5500-X with Inside/Outside/DMZ zones
Network Segmentation	VLANs 10/20/50/70/90/199 with inter-VLAN ACLs
DMZ	Public servers isolated in 10.11.11.0/27 DMZ segment
NAT/PAT	Internal RFC 1918 addresses hidden behind public ISP IPs
Access Control	ACLs on firewall restrict traffic by protocol and direction
SSH Access Control	Standard ACL restricts SSH to Management VLAN (192.168.10.0/24) only

Security Control	Implementation
Blackhole VLAN	VLAN 199 assigned to all unused switch ports
STP Security	PortFast + BPDU Guard on all access ports
Authentication	RADIUS server for 802.1X wireless and device authentication
Password Security	All device passwords encrypted (service password-encryption)
Banner	MOTD banner: 'NO UNAUTHORIZED ACCESS' on all devices
Gateway Redundancy	HSRP on all core SVIs — automatic failover < 10 seconds
WAN Redundancy	Dual ISPs with floating static routes (metrics 1 and 70)

14. Testing & Verification Results

14.1 Connectivity Tests

Test	Method	Result
Intra-VLAN Connectivity	Ping between hosts in same VLAN	PASS
Inter-VLAN Routing	Ping between VLANs via core switch SVIs	PASS
DHCP Address Assignment	IPCONFIG on clients in all departments	PASS
Default Gateway Reachability	Ping HSRP virtual IP from all VLANs	PASS
Internal to DMZ	Ping from LAN to DMZ server IPs	PASS
Internal to Internet	Ping 8.0.0.20 (simulated Indian/Russian user)	PASS
OSPF Full Adjacency	show ip ospf neighbor — all neighbors FULL	PASS
HSRP Failover	Disable Active switch; verify Standby takes over	PASS
EtherChannel LACP	show etherchannel summary — Po1(SU) with 3 ports	PASS
Dual-ISP Failover	Shut ISP-1 link; verify traffic via ISP-2	PASS
SSH Access Restriction	SSH from non-MGT VLAN — rejected by ACL	PASS
WLC + LAP Registration	LAPs associated to WLC; clients obtain WLAN IP	PASS
VoIP Calls	IP phones register; calls established between departments	PASS
NAT/PAT to Internet	Internal hosts access external IP 8.0.0.0/8	PASS

14.2 OSPF Verification Summary

Core-SW1 maintained 6 OSPF full adjacencies — to both firewalls (via routed uplinks) and to Core-SW2 (via all VLAN SVIs). All adjacencies reached FULL state, confirming complete link-state database synchronization across the enterprise.

15. Conclusion

The Cloud Company Secure Network Design project has been successfully designed, configured, and verified. The implemented infrastructure meets all specified requirements:

- 600-user capacity across three floors, five departments, and one ICT/Server floor
- Hierarchical design with Core Multilayer Switches and departmental Access Switches
- Full VLAN segmentation (Management, LAN, WLAN, VoIP, Inside Server, Blackhole)
- Dual-ISP redundancy with automatic failover via floating static routes
- Cisco ASA firewalls enforcing security zones with DMZ for public-facing services
- HSRP providing transparent gateway redundancy across all VLANs
- LACP EtherChannel delivering 3 Gbps aggregate bandwidth between core switches
- OSPF providing dynamic, loop-free routing with fast convergence
- Centralized wireless management via Cisco WLC with RADIUS authentication
- VoIP services with traffic segregation on dedicated VLAN
- Comprehensive security controls: ACLs, NAT, STP security, SSH restrictions

All components have been tested and verified through comprehensive simulation in Cisco Packet Tracer. The network is production-ready and positioned to scale with Cloud Company's continued growth.

— End of Document —